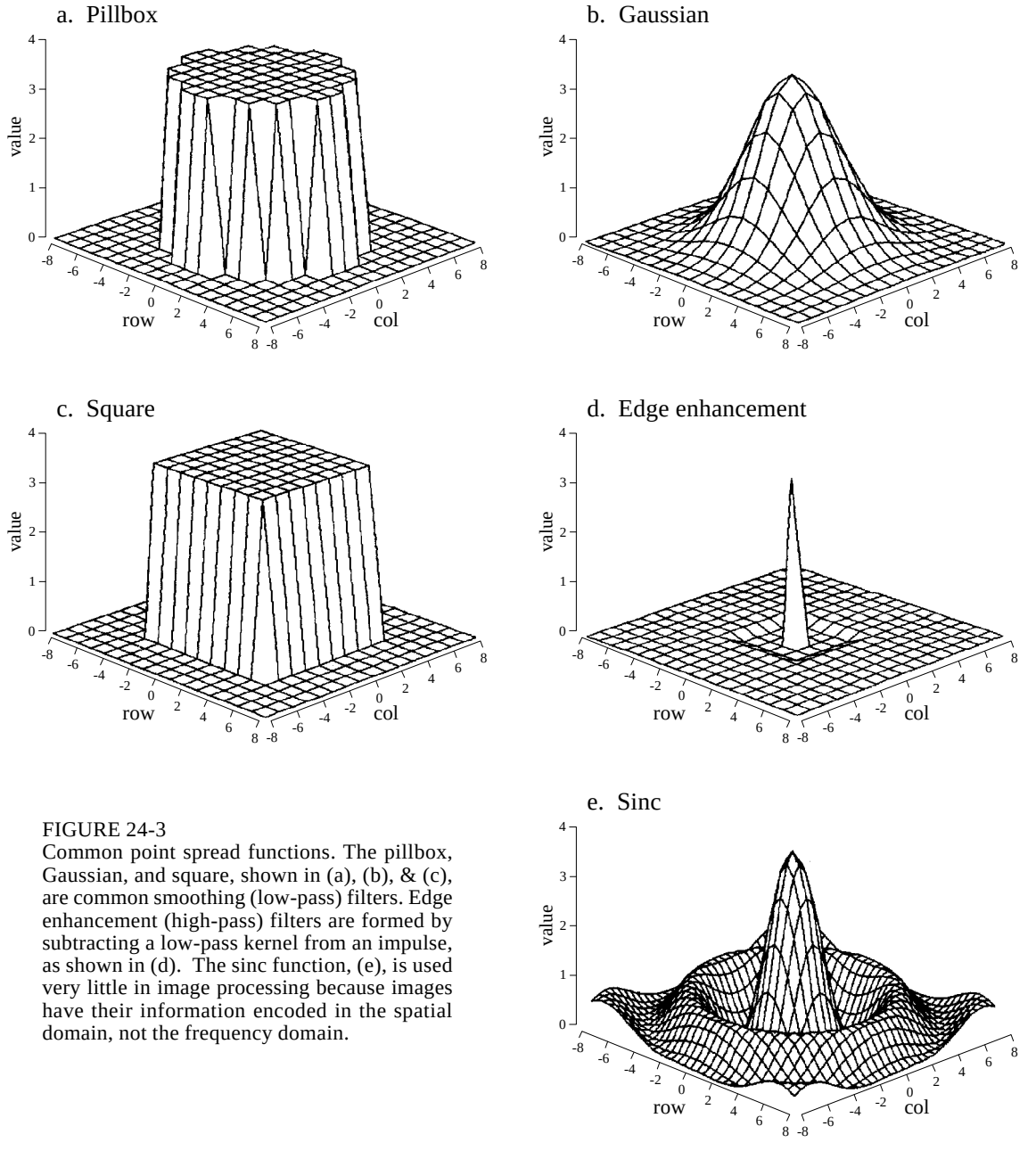

FIGURE 24-3
Common point spread functions. The pillbox, Gaussian, and square, shown in (a), (b), & (c), are common smoothing (low-pass) filters. Edge enhancement (high-pass) filters are formed by subtracting a low-pass kernel from an impulse, as shown in (d). The sinc function, (e), is used very little in image processing because images have their information encoded in the spatial domain, not the frequency domain.

All these filter kernels use *negative* indexes in the rows and columns, allowing the PSF to be centered at *row* = 0 and *column* = 0. Negative indexes are often eliminated in one-dimensional DSP by shifting the filter kernel to the right until all the nonzero samples have a positive index. This shift moves the output signal by an equal amount, which is usually of no concern. In comparison, a shift between the input and output images is generally not acceptable. Correspondingly, negative indexes are the norm for filter kernels in image processing.